

data bits. Each address/data input can be set to one of the two logic states. Upon receipt of a trigger sig-

nal, the programmed addresses/data are transmitted together with header bits via an RF transmission medium.

The capability to select a TE trigger on the HT12E further enhances the application flexibility of 2^{12} series of encoders. HT12E IC has 8 address bits and 4 data bits.

HT12D decoder. This 2^{12} series CMOS LSI decoder is widely used with HT12E encoder for remote control applications. It

receives serial addresses and data transmitted from the programmed 2^{12} series encoder. The decoder compares the serial input data three times continuously with its local addresses. If no error or unmatched code is found, input data codes are decoded and then transferred to the output pins.

The 2^{12} series of decoders are capable of decoding the information comprising 'N' bits of address and '12-N' bits of data. The HT12D is arranged to provide 8 address bits and 4 data bits. For a valid transmission, address bits of both the encoder and the decoder must be the same.

SPDT relay. Relays used here are single-pole double-throw (SPDT) type. In this type of relay, the output side has three terminals: normally open (N/O), normally closed (N/C) and common (C).

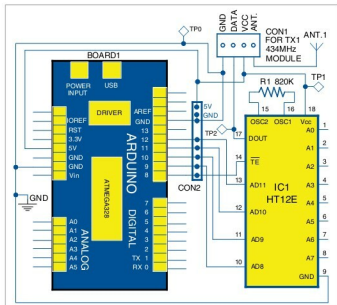


Fig. 5: Circuit diagram of the transmitter unit

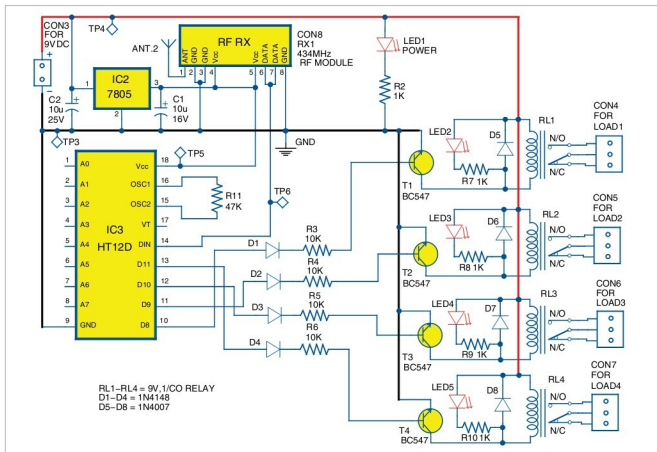


Fig. 6: Circuit diagram of the receiver unit